

Efficient Video Object Segmentation and Tracking with Recurrent Dynamic Submodel

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Abstract

*Large vision foundation models, such as SAM2, have achieved remarkable performance in video object segmentation and tracking (VOST). However, their effectiveness is hindered by significant computational overhead. While model pruning is a widely used strategy to address this issue, traditional static and input-agnostic pruning approaches fall short in managing the diverse and complex nature of video data effectively. A promising alternative is dynamic networks, yet they often struggle to translate theoretical computational reductions into actual acceleration. Furthermore, both static and dynamic approaches typically focus on visual features of individual frames while neglecting the temporal correlations between them, limiting their performance in handling complex video streams. To address these challenges, we propose **Recurrent Dynamic Submodel (RDS)**, a dynamic architecture that adaptively selects submodel blocks for each frame. Specifically, it has a lightweight Prediction-Aware Router (PAR), which leverages both the segmentation mask from the previous frame and the visual features of the current frame to make routing decisions, enabling the submodel to explicitly capture the temporal nature of video data. Additionally, to reduce the cost of adapting the dynamic submodel, we introduce an Importance-aware LoRA (I-LoRA), tuning parameters only in the most critical blocks. Extensive experiments show that our approach achieves an optimal trade-off between performance and speed, requiring minimal trainable parameters and training data.*

1. Introduction

Recent advancements in large vision foundation models, such as SAM2 [32], have significantly propelled progress in video object segmentation and tracking (VOST) [3, 8, 23, 42, 46, 50, 52]. While these models demonstrate remark-

able performance, their broader application remains constrained by their substantial computational requirements, particularly in resource-limited environments.

To tackle this challenge, prior works [4, 11] have proposed static model pruning techniques to reduce model size. However, these input-agnostic approaches often struggle with temporally varied and complex video data. This is because different tasks, datasets, or even individual video frames may favor distinct key components of the original model, as illustrated in Figure 1 (a).

Dynamic models [17, 26, 36, 38] offer an effective alternative to static model pruning by adaptively activating a submodel customized for each input frame. This dynamic activation is typically implemented using a lightweight router, which conditions on intermediate features to selectively activate components along the computational path, such as tokens [21, 25, 26, 30, 31, 47, 51] and blocks [2, 13–15, 17, 20, 26, 35, 45]. However, while they achieve theoretical FLOPs reduction, the acceleration benefits are often offset by the additional overhead introduced by per-block routers, as illustrated in Figure 1 (b).

Moreover, both the pruning criteria used in static model pruning approaches and the intermediate features employed for routing in dynamic models are often adapted from image-based models by treating video streams as mere sequences of individual images. This simplification generally overlooks the temporal correlations between consecutive frames, thereby limiting their performance potential.

To address the aforementioned challenges, we propose the **Recurrent Dynamic Submodel (RDS)**, a dynamic architecture designed for efficient VOST. Our framework adaptively activates a subset of blocks for each video frame, guided by a lightweight, prediction-aware router. This router leverages the segmentation mask of the previous frame as a temporal prior and combines it with visual features from the current frame. By fusing temporal and visual information, the router adaptively determines the activation of submodels, effectively overcoming the temporal-agnostic limitations of prior approaches. Unlike existing

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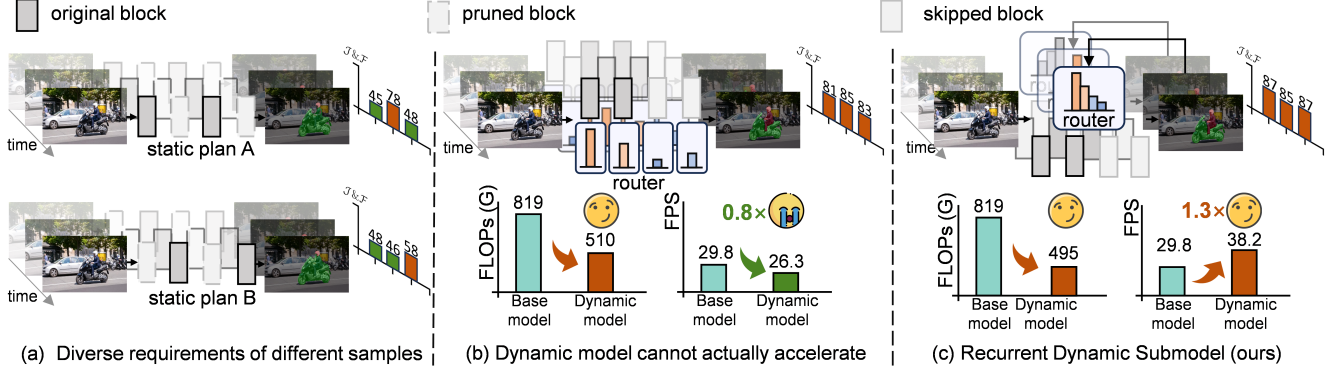


Figure 1. (a) **Different samples favor different submodel architectures.** Two static models, obtained by randomly pruning at the same rate, achieve similar overall results on the dataset but exhibit significant variations in their frame-by-frame performance. (b) **The per-block router in dynamic models fails to deliver real acceleration.** We employ the block-skipping mechanism in AdaViT [26], which includes a router for each block. This approach results in a theoretical reduction in FLOPs but causes a slowdown in practice. (c) **Our recurrent dynamic submodel (RDS)** selects a subset of blocks for each frame using a single router, leveraging the prediction from the previous frame, achieving both FLOPs reduction and actual acceleration.

methods that assign a dedicated router to each block for dynamic activation, our approach employs a single router to determine the subset of active blocks for the entire model. This significantly reduces the overhead associated with dynamic routing, leading to both reduced computational costs and speedup in practice, as illustrated in Figure 1 (c).

The dynamic activation decision actually reflects an insight that the importance of blocks is different and there may exist a core subset of blocks that are fundamentally more important across diverse samples. Motivated by this observation, we propose an importance-aware low-rank adaptation (I-LoRA) method, an efficient training approach that updates parameters exclusively within these important blocks. Specifically, it employs a dual-perspective importance metric, evaluating each block’s significance based on its impact on the final output and its influence on the current feature map. This design complements RDS, enabling a substantial reduction in the adaptation costs of our dynamic framework while preserving competitive performance.

In summary, our contributions are summarized as:

- We propose the **Recurrent Dynamic Submodel (RDS)** framework. Its prediction-aware router adopts previous predictions and current inputs to dynamically select a subset of blocks per frame, overcoming the input-agnostic nature of static approaches and eliminating the overhead of per-block routers in dynamic methods.
- We introduce the I-LoRA strategy, which employs a dual importance metric to identify the most vital blocks, then selectively allocates trainable parameters to these blocks, thus significantly lowering adaptation costs.
- Extensive empirical validation on multiple challenging VOS and VOT benchmarks demonstrates our method achieves significant real-world acceleration. For example, it achieves a **1.3×** speedup on the DAVIS 2017 [28]

dataset with less than **0.4%** performance degradation, while introducing only **3%** trainable parameters and requiring minimal SAM2 training data.

- Additional experiments on DINOv3 [33] further demonstrate the generalizability of our framework across different vision foundation models.

2. Related Work

2.1. Video Object Segmentation and Tracking

Video Object Segmentation (VOS) [8, 23, 46, 52] and Video Object Tracking (VOT) [3, 46, 50] are fundamental and closely related areas in computer vision. While VOS aims for pixel-level target segmentation and VOT focuses on bounding-box localization, both require robust identity modeling across frames to handle challenges like occlusion and deformation.

Driven by recent advancements, state-of-the-art methods in both fields increasingly rely on large foundation models [9, 32, 52]. Despite their eminent zero-shot generalization and tracking performance, these heavy architectures incur prohibitive computational overhead, hindering their deployment in real-time video applications. Although recent efforts have actively explored accelerating and lightening these foundation models (e.g., via knowledge distillation or efficient network designs) [5, 40, 41, 49], balancing performance and efficiency remains a critical bottleneck. Crucially, these lightweight variants often employ static architectures, meaning they allocate uniform computational resources to every frame. This ignores the inherent temporal redundancy and varying complexity of video streams.

To address this, we propose Recurrent Dynamic Submodel, which achieves faster inference and lower adaptation cost by dynamically skipping encoder blocks, fine-tuning only LoRA adapters assigned to critical blocks.

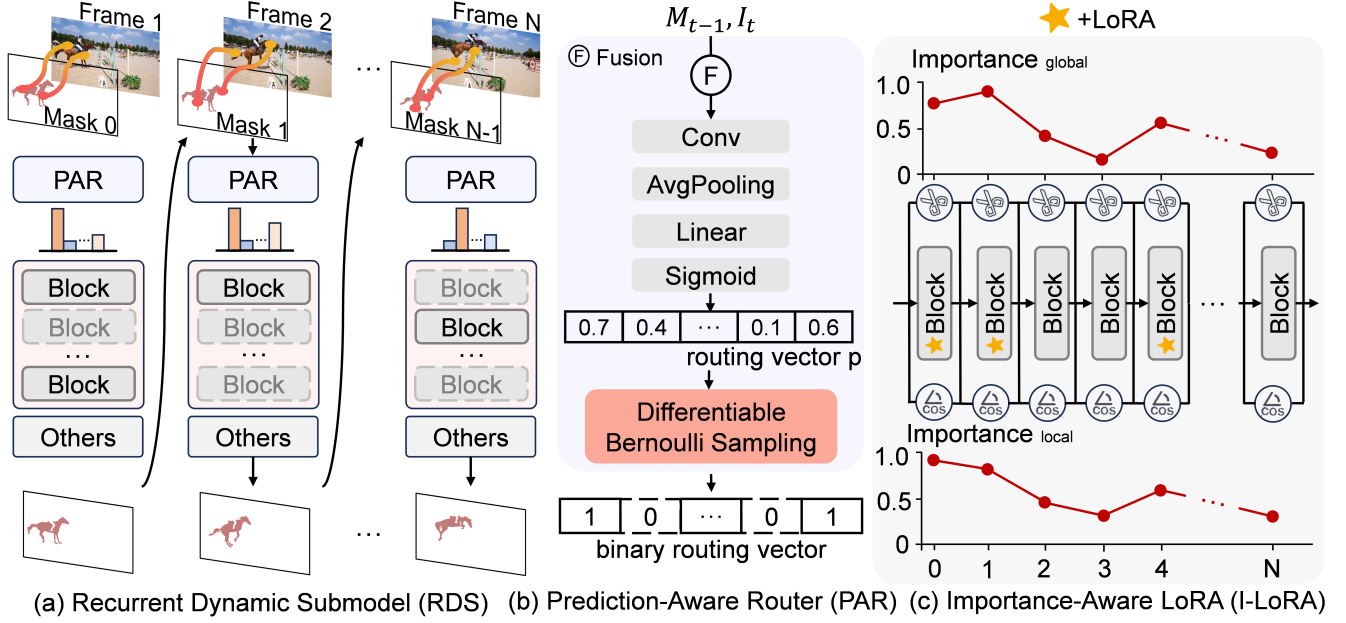


Figure 2. (a) **The overall framework of RDS.** It dynamically activates a submodel for each frame. (b) **Prediction-Aware Router (PAR)** takes the fused feature from the current frame’s visual features and the previous frame’s segmentation mask as input and outputs a binary routing vector for block selection. (c) **Importance-aware LoRA (I-LoRA)** identifies critical blocks using both global (impact on final output) and local (feature transformation) importance. Trainable LoRAs are allocated only to these key blocks for efficient fine-tuning.

2.2. Dynamic Models

Dynamic models [10, 12, 17, 26, 34, 36–38] selectively activate only part of their computational pathways (*e.g.*, tokens [21, 25, 26, 30, 31, 47, 51] or blocks [2, 13–15, 17, 20, 26, 35, 45, 48]) to achieve the goal of computational efficiency, given intermediate features or task complexity as conditions. Unlike traditional static models, these networks adaptively adjust their components based on different input data, demonstrating strong performance in tasks such as image recognition [2, 30] and object detection [14]. As for visual segmentation, existing research [26] has integrated dynamic mechanisms into encoder architectures, enhancing inference efficiency by skipping redundant modules. In addition, there are studies [24] that leverage redundancy between consecutive frames, avoiding unnecessary computation, and reducing latency.

Still, current approaches face challenges in balancing accuracy and efficiency, characterized by their dense routing, *i.e.*, allocating routers for intermediate features from all stages. This can introduce repetitive computations and hinder the actual accelerating effect.

We address these limitations by introducing a Prediction-Aware Router that jointly takes temporal and visual features as input, reaching a dynamic activation decision with a single routing computation. Additionally, we propose I-LoRA to enable efficient fine-tuning of dynamic architectures while maintaining performance.

3. Recurrent Submodel Dynamics

Our approach, illustrated in Figure 2, introduces two core components. First, a Prediction-Aware Router (PAR) dynamically selects a submodel for each frame using visual features and the previous frame’s mask. Second, Importance-Aware LoRA (I-LoRA) enables efficient fine-tuning by adapting only the most critical blocks. We detail the full training and inference process in Section 3.4.

3.1. Preliminary

Segment Anything Model 2 (SAM2). This section briefly introduces the Segment Anything Model 2 (SAM2) [32], a representative vision foundation model for VOST. Its transformer-based encoder first maps image pixels $\mathbf{I}$ to embeddings $\mathbf{X}$ via a patch embedding layer P , *i.e.* $\mathbf{X} = P(\mathbf{I})$. These embeddings are then passed through cascading transformer blocks, each comprising a multi-head self-attention (MHSA) mechanism and a multi-layer perceptron (MLP). The operations can be formulated:

$$\text{MHSA}(\mathbf{X}) = \text{Concat}(\text{head}_1, \dots, \text{head}_H) \mathbf{W}^O,$$

$$\text{where } \text{head}_h = \text{Attention}(\mathbf{X} \mathbf{W}_h^Q, \mathbf{X} \mathbf{W}_h^K, \mathbf{X} \mathbf{W}_h^V); \quad (1)$$

$$\text{MLP}(\mathbf{X}) = \sigma(\mathbf{X} \mathbf{W}^U) \mathbf{W}^D.$$

where $\mathbf{W}_h^Q$, $\mathbf{W}_h^K$, $\mathbf{W}_h^V$ are the query, key, and value projection matrices for the h -th attention head, $\mathbf{W}^O$ is the output projection matrix for MHSA, and $\mathbf{W}^U$, $\mathbf{W}^D$ are the up- and down-projection matrices in the MLP.

SAM 2 incorporates a sophisticated memory mechanism where the segmentation prediction of the current frame is conditioned on memories of past frames. To simplify the discussion, we consider only the consecutive preceding frame. Given the current frame $\mathbf{I}_t$, the previous segmentation result $\mathbf{M}_{t-1}$, encoder E , decoder D , the memory mechanism can be formulated as:

$$\mathbf{M}_t = D(E(\mathbf{I}_t, \mathbf{M}_{t-1})), \quad (2)$$

Although this memory mechanism is widely adopted in VOST, existing efficient solutions, particularly concerning dynamic models, often overlook this valuable temporal prior when performing dynamic activation. In contrast, our method leverages this temporal information by explicitly incorporating the previous segmentation mask $\mathbf{M}_{t-1}$ as a key input to our router.

Low-Rank Adaptation. LoRA [16] is introduced to significantly reduce the overhead of model training by injecting lightweight, trainable adapters into existing layers while keeping the original model weights frozen. The updated weight $\mathbf{W}'$ can be formulated as:

$$\mathbf{W}' = \mathbf{W}_0 + \mathbf{B}\mathbf{A}, \quad (3)$$

where $\mathbf{W}_0$ denotes the original weight matrix, $\mathbf{B} \in \mathbb{R}^{d \times r}$ and $\mathbf{A} \in \mathbb{R}^{r \times k}$ are low-rank matrices with $r \ll \min(d, k)$. The objective is to minimize the training by only updating the low-rank components and freezing the original weight.

Our I-LoRA further reduces the trainable parameters in our model by identifying the most critical blocks across diverse samples and inserts LoRA adapters into the weights of these blocks, including $\mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V, \mathbf{W}^O$ in the MHSA block, as well as $\mathbf{W}^U$ and $\mathbf{W}^D$ in the MLP block.

3.2. Prediction-Aware Router

The dynamic block selection in our approach relies on a prediction-aware router. This router takes both visual features of the current frame and the segmentation mask of the previous frame as inputs, fuses these inputs, and outputs the probability of activating each block eventually.

Specifically, given the patchified embedding of an input frame $\mathbf{X}_t \in \mathbb{R}^{H \times W \times C}$ and segmentation mask of the previous frame $\mathbf{M}_{t-1} \in \{0, 1\}^{H \times W \times 1}$, we fuse them via:

$$\mathbf{Y}_t = F(\mathbf{X}_t, \mathbf{M}_{t-1}) = \alpha \cdot f(\mathbf{M}_{t-1}) + g(\mathbf{X}_t), \quad (4)$$

where F is the fusing function, f and g are different convolution layers that align hidden dimensions and extract information, t is the temporal index of the frame, α is a learnable scalar, and $\mathbf{Y}_t$ is the fused feature.

Next, $\mathbf{Y}_t$ is passed to the router ϕ consisting of a convolution, average pooling, and a linear projection, resulting in the raw logits $\mathbf{l}_t \in \mathbb{R}^L$:

$$\mathbf{l}_t = \phi(\mathbf{Y}_t), \quad (5)$$

To enable end-to-end optimization of discrete routing, we adopt a Straight-Through Estimator (STE) with Bernoulli sampling [1, 17]. Given routing logits $\mathbf{l}_t$, we apply a sigmoid function followed by a normalization step controlled by a target activation ratio $\beta \in (0, 1]$, yielding the continuous probabilities $\tilde{w}_t^i$ for each block:

$$\tilde{w}_t^i = \min \left(\frac{\beta \cdot L \cdot \sigma(l_t^i)}{\sum_{j=1}^L \sigma(l_t^j)}, 1.0 \right), \quad (6)$$

where L is the total number of blocks (e.g., $L = 48$ in SAM2). This normalization inherently bounds the expected number of activated blocks.

To ensure basic feature extraction and maintain model stability, a predefined subset of critical blocks (denoted as $\mathcal{K}$) is permanently kept active regardless of the router's prediction. For the remaining blocks, the binary routing decision r_t^i during training is generated via Bernoulli sampling based on the normalized probability $\tilde{w}_t^i$. The routing state for the i -th block is defined as:

$$r_t^i = \begin{cases} 1, & \text{if } i \in \mathcal{K}, \\ \tilde{r}_t^i, & \tilde{r}_t^i \sim \text{Bernoulli}(\tilde{w}_t^i), \text{ otherwise.} \end{cases} \quad (7)$$

Given the i -th hidden state $\mathbf{X}_t^i$ and encoder block h^i , the feature propagation is computed as:

$$\mathbf{X}_t^{i+1} = r_t^i \cdot h^i(\mathbf{X}_t^i) + (1 - r_t^i) \cdot \mathbf{X}_t^i \quad (0 \leq i \leq L-1). \quad (8)$$

To handle the non-differentiability of sampling, we adopt STE in backpropagation, treating the discrete step as identity, i.e., $\frac{\partial \mathcal{L}}{\partial \tilde{w}_t^i} \approx \frac{\partial \mathcal{L}}{\partial r_t^i}$ where $\mathcal{L}$ is SAM2 loss. The evaluation protocol is described in Section 3.4, where stochastic evaluation is used by default and deterministic thresholding is treated as a deployment approximation.

3.3. Importance-Aware LoRA

Dynamic activation reveals that only a subset of blocks is needed for some samples. This implies the potential existence of an uneven importance distribution. We thus propose the Importance-aware Low-Rank Adaptation (I-LoRA) to further alleviate the training burden of adapting the dynamic architecture. This approach identifies the most critical blocks from both global and local perspectives, and applies LoRA only to them. Details are presented below:

Global perspective. The global importance of the i -th block $\mathcal{I}_G^i$ reflects its contribution to the final output by measuring the change in $\mathcal{J}\&\mathcal{F}$ score [28] upon its removal:

$$\mathcal{I}_G^i = \mathbb{E} [\mathcal{J}\&\mathcal{F}(\mathcal{O}) - \mathcal{J}\&\mathcal{F}(\mathcal{O}^i)], \quad (9)$$

where $\mathcal{O}$ and $\mathcal{O}^i$ are the outputs of the original model with/without the i -th block, $\mathcal{J}\&\mathcal{F}(\cdot)$ is calculated with the ground truth.

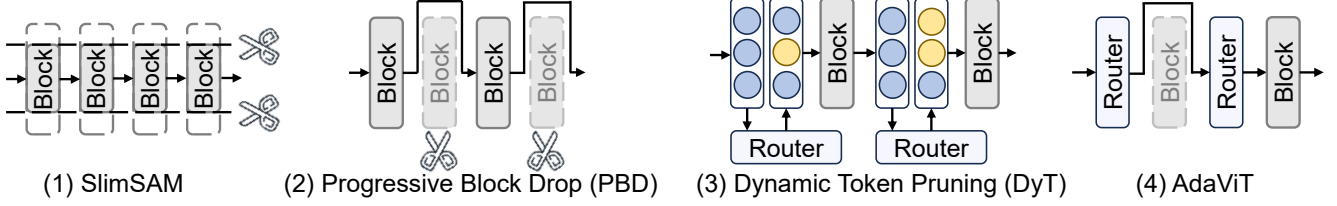


Figure 3. Illustration of the four baseline methods. **(1) SlimSAM** [5]: A static channel pruning method that permanently removes network channels. **(2) PBD** [4]: A static block pruning strategy that removes entire computational blocks. **(3) DyT** [51]: A dynamic token pruning method where blue circles represent kept tokens and yellow circles represent skipped tokens. **(4) AdaViT** [26]: A dynamic block skipping method that decides whether to execute certain blocks.

Local perspective. The local importance $\mathcal{I}_L^i$ indicates the block’s contribution to feature transformation. A significant transformation suggests the block is performing a critical computation. We measure this via the cosine similarity between the input and output features of the block:

$$\mathcal{I}_L^i = 1 - \mathbb{E} \left[\frac{\langle \mathbf{X}^i, \mathbf{X}^{i+1} \rangle}{\|\mathbf{X}^i\| \cdot \|\mathbf{X}^{i+1}\|} \right], \quad (10)$$

where $\langle \cdot, \cdot \rangle$ and $\|\cdot\|$ are inner product and L2-norm operation of certain vector, respectively.

Parameter allocation. We select adapted blocks via rank aggregation over global and local importance. Let K_{sel} denote the target number of blocks to fine-tune. For each $i \in \{0, \dots, L-1\} \setminus \mathcal{K}$, we compute the descending ranks $r_G(i)$ and $r_L(i)$ from $\mathcal{I}_G^i$ and $\mathcal{I}_L^i$, and define:

$$s(i) = -(r_G(i) + r_L(i)).$$

The final selected set of blocks $\mathcal{H}$ is:

$$\mathcal{H} = \mathcal{K} \cup \text{TopIdx}_{K_{\text{sel}} - |\mathcal{K}|}(s). \quad (11)$$

where $\text{TopIdx}_k(\cdot)$ returns the indices of the top k values. For each block in $\mathcal{H}$, we inject LoRA adapters into all projection matrices, namely $\mathbf{W}^Q$, $\mathbf{W}^K$, $\mathbf{W}^V$, $\mathbf{W}^O$, $\mathbf{W}^U$, and $\mathbf{W}^D$, optimizing only these adapters and the router while freezing the original encoder weights.

3.4. Training and Inference of RDS

Training. Our training pipeline can be roughly divided into 2 stages: i) I-LoRA identification and ii) Dynamic components fine-tuning. Specifically, we first split off a small subset of the training set and run inference on these data using the SAM2 backbone, locating the most important blocks as in Equation 11. LoRA adapters are then inserted into all weight matrices of these blocks following Equation 3.

After we have allocated trainable parameters, we fine-tune our router and the LoRA adapters jointly. Unlike traditional dynamic methods that rely heavily on a delicate sparsity loss penalty, our normalized routing probability $\tilde{w}_t^i$ (as defined in Equation 6) inherently constrains the overall activation ratio to approximately β . During training, we fix the hyperparameter β to strictly control the target complexity.

Inference. We report stochastic evaluation by default: for $i \notin \mathcal{K}$, we sample $r_t^i \sim \text{Bernoulli}(\tilde{w}_t^i)$, which better matches the training objective. For deterministic deployment, we instead set $r_t^i = 1$ when $\tilde{w}_t^i \geq \tau$ and $r_t^i = 0$ otherwise, with τ calibrated on a validation set (default $\tau = 0.5$); blocks in $\mathcal{K}$ remain always active.

4. Experiments

4.1. Experiment Settings

Our experiments cover two tasks: Video Object Segmentation (VOS) and Video Object Tracking (VOT).

Datasets. For the VOS task, the model is evaluated in a semi-supervised setting, where the ground-truth mask for the first frame is provided. We adopt three main datasets for this evaluation: DAVIS 2017 [28], SA-V [32], and YouTube-VOS (YTVOS 2019) [43]. For VOT experiments, we evaluate our method using the following datasets: LaSOT [7], TrackingNet [27], and GOT-10k [18].

Implementation Details. We fine-tune the model on DAVIS 2017 [28] with I-LoRA for 40 epochs on a single RTX A5000 ADA, following SAM2 fine-tuning for other settings. I-LoRA employs rank 32, applied to half of encoder blocks (*i.e.*, 24 blocks) selected via the importance metric. For evaluation, we adopt semi-supervised VOS setting, where ground-truth masks are provided for the first frame.

The baseline and default settings in the experiments are highlighted using **grey** and **blue**, respectively.

4.2. Comparison with Other Methods

As shown in Figure 3, we compare RDS against four mainstream strategies (SlimSAM [5], Progressive Block Drop [4], Dynamic Token Pruning [51], and AdaViT [26]), all based on the SAM2 backbone. Further details on these four methods are provided in the supplementary material.

As shown in Table 1 and Figure 4, RDS lies on the Pareto frontier, offering a strong trade-off at high performance. For instance, on DAVIS 2017, it delivers a notable **1.3** $\times$ speedup with negligible performance cost, significantly outperforming others. Unlike methods like DyT and AdaViT,

Table 1. **Performance comparison with other methods.** This table compares the performance of different methods in terms of trainable parameters, training set size, GPU usage, number of iterations, FLOPs, FPS, and $\mathcal{J}\&\mathcal{F}$ score [28] on DAVIS 2017.

Method	Trainable param. (M)	Train set size (%)	GPUs	Iter. (K)	FLOPs (G)↓	FPS ↑	$\mathcal{J}\&\mathcal{F}$ ↑
SAM2	224	100	256	200	819	29.8	90.0
SlimSAM	147	>5	8	38	547	33.4	87.8
PBD	7.6	<0.03	1	5	500	39.8	85.6
DyT	20.5	<0.03	1	5	629	22.3	90.2
AdaViT	9.6	<0.03	1	5	528	26.3	88.3
RDS (Ours)	7.6	<0.03	1	5	500	<u>38.2</u>	<u>89.6</u>

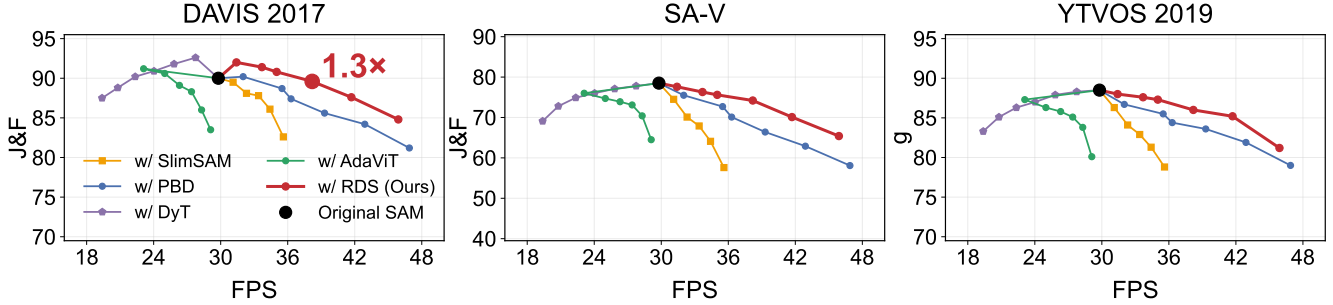


Figure 4. **Our RDS method consistently leads in the performance-efficiency trade-off.** This figure compares the performance ($\mathcal{J}\&\mathcal{F}$ score [28]) versus practical inference speed (FPS) of different methods on the DAVIS 2017 [28], SA-V [32], and YTVOS 2019 [43] validation sets. RDS achieves significant real-world acceleration (e.g., $1.3\times$ on DAVIS 2017) while maintaining high accuracy, whereas other dynamic methods show limited or even negative speedup due to routing overhead.

Table 2. **Evaluation of the SAMURAI model on VOT benchmarks.** This table presents the AUC and FPS metrics for the SAMURAI model and its variants on the LaSOT, TrackingNet, and GOT-10k datasets.

Method	AUC			FPS
	LaSOT	TrackingNet	GOT-10k	
SAMURAI	75.3	85.3	84.1	28.5
w/ RDS _{0.7}	72.9	83.1	82.3	33.2 (1.2 $\times$)
w/ RDS _{0.6}	72.0	82.6	82.0	35.7 (1.3 $\times$)
w/ RDS _{0.5}	70.9	81.2	80.9	38.6 (1.4 $\times$)

which are hindered by dense routing overhead, the single, lightweight router design of RDS successfully converts computational savings into real-world acceleration.

4.3. Main results

To demonstrate cross-domain transferability, we exclusively fine-tune the dynamic components (*i.e.*, router and LoRA adapters) of our SAM2-Hiera-L baseline on DAVIS 2017 dataset. These are directly integrated into SAMURAI [44], SAM2Long [6], and SAM2 [32] without task-specific retraining. This minimal-cost setup highlights RDS’s strong generalization, inviting the community to explore broader dataset fine-tuning.

From Table 2 and Table 3, we can conclude several insights: i) RDS preserves most of the original models’ per-

Table 3. **Evaluation of different models on VOS benchmarks.** This table shows the $\mathcal{J}\&\mathcal{F}$, $\mathcal{G}$, and FPS metrics for the SAMURAI, SAM2Long, and SAM2 models and their variants on the DAVIS 2017, SA-V, and YTVOS 2019 datasets.

Method	$\mathcal{J}\&\mathcal{F}$		$\mathcal{G}$	FPS
	DAVIS 2017	SA-V		
SAMURAI	90.1	78.8	88.3	28.5
w/ RDS _{0.7}	90.8	76.1	86.8	33.2 (1.2 $\times$)
w/ RDS _{0.6}	90.1	74.7	85.6	35.7 (1.3 $\times$)
w/ RDS _{0.5}	87.8	72.4	85.2	38.6 (1.4 $\times$)
SAM2Long	90.2	81.0	88.4	15.8
w/ RDS _{0.7}	90.6	78.1	87.4	17.5 (1.1 $\times$)
w/ RDS _{0.6}	90.0	76.8	86.7	18.4 (1.2 $\times$)
w/ RDS _{0.5}	87.9	74.6	85.6	19.8 (1.3 $\times$)
SAM2	90.0	78.5	88.5	29.8
w/ RDS _{0.7}	90.8	75.6	87.3	35.1 (1.2 $\times$)
w/ RDS _{0.6}	89.6	74.3	86.0	38.2 (1.3 $\times$)
w/ RDS _{0.5}	87.6	70.1	85.2	41.5 (1.4 $\times$)

formance across various tasks and methods; this trend remains even when the block skip rate increases. ii) RDS provides real-world acceleration with strong transferability while maintaining performance. Specifically, on DAVIS 2017 with SAM2, it achieves a $1.3\times$ speedup with a perfor-



Figure 5. **Visual comparison on the DAVIS 2017 dataset.** The bottom row shows the original SAM2 model, and the top row shows our RDS model (activation ratio 0.6). The results demonstrate that our method maintains high-quality segmentation, nearly indistinguishable from the original model, while significantly improving computational efficiency.

mance drop of less than **0.4%**.

4.4. Ablation Study

To validate the effectiveness of the key components within our proposed framework, we conduct a series of comprehensive ablation studies. Unless otherwise specified, all experiments are conducted on the DAVIS 2017, with a visual comparison provided in Figure 5.

Table 4. **Routing latency comparison.** A single router determines the architecture from the input, reducing routing latency from ~ 17.2 ms to ~ 0.5 ms and enabling real speedup.

Strategy	Total Latency	Block Latency	Router Latency
Original SAM2	~ 23.1 ms	~ 23.1 ms	-
SAM2 w/ Dense Router	~ 31.1 ms	~ 13.9 ms	~ 17.2 ms
SAM2 w/ Single Router	~ 14.5 ms	~ 13.9 ms	~ 0.5 ms

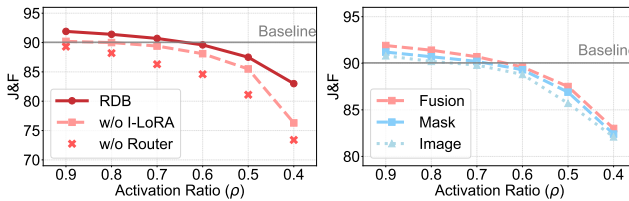


Figure 6. **The contributions of key components in RDS:** put: Fusion of temporal prior and visual features achieves the best performance. The prediction-aware router and Importance-aware LoRA.

Effectiveness of the Router. Unlike dense per-block routing, our single router predicts the active submodel directly from the input, reducing latency from 17.2 ms to 0.5

ms (Table 4). Moreover, it outperforms static pruning under the same budget (Figure 6), highlighting the need for adaptive submodels.

Impact of Input on Routing Decisions. Our router fuses current visual features with a temporal prior (previous mask). Figure 7 shows that image-only routing struggles with background distractions, while mask-only routing fails to adapt to appearance changes. By synergizing spatial attention (where to look) with visual cues (what to see), our fused approach achieves the best performance.

Table 5. **Router Memory.** The immediate prior mask (M_{t-1}) suffices for routing.

Memory	$\mathcal{J}\&\mathcal{F}$
M_{t-1}	89.6
$M_{t-1} \cup M_{t-2}$	89.6
$M_{t-1} \dots \cup M_{t-4}$	89.4

Table 6. **Motion Bias.** Minimal degradation under large inter-frame motion (low FPS).

Downsample ratio	$\mathcal{J}\&\mathcal{F}$
$1 \times (\sim 24\text{FPS})$	89.6
$4 \times (\sim 6\text{FPS})$	88.8
$6 \times (\sim 4\text{FPS})$	88.7

Temporal information. As shown in Table 5, regarding the memory utilized by the router, extending the history masks ($M_{t-1} \cup \dots$) yields no noticeable performance improvement. This further demonstrates that using only the most recent prior mask M_{t-1} is sufficient to support effective and stable dynamic routing.

Motion/FPS Bias. In Table 6, we downsample videos up to $6 \times (\sim 4 \text{ FPS})$ to simulate large inter-frame motion. The minimal performance degradation confirms the robustness of using the previous frame’s mask as a temporal prior, even under varied motion scales and large object displacements.

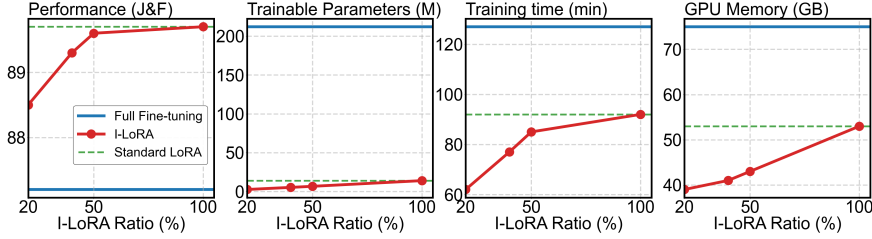


Figure 8. **I-LoRA vs Baselines.** Comparison of our I-LoRA (at various ratios) with standard LoRA and full fine-tuning. I-LoRA matches or exceeds performance while significantly reducing training time and GPU memory.

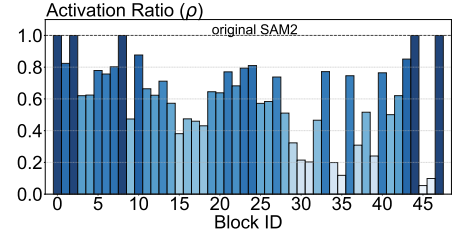


Figure 9. **Average activation rate per encoder block on DAVIS 2017.** Activation is selective and largely depth-independent.

Effectiveness of the I-LoRA Metric. We evaluate strategies for allocating parameters to critical blocks. As shown in Table 7, using only local or global importance is suboptimal, while I-LoRA performs best. For global importance, we compare *per block* and *per block-group*. The latter reduces time cost by $4\times$ with minimal loss, achieving a strong efficiency–accuracy trade-off.

Table 7. **Effectiveness of I-LoRA Parameter Allocation.** The dual-perspective strategy consistently outperforms single-perspective baselines. Using the block-group strategy for global importance also provides a $4\times$ speedup over per-block with minimal performance loss.

Local-Only	Global-Only (per block)	Global-Only (per block-group)	Time Cost	$\mathcal{J}\&\mathcal{F}$
✓			~4 mins	88.7
	✓		~3 h	89.2
		✓	~44 mins	89.0
✓	✓		~3 h	89.6
✓		✓	~48 mins ($4\times$)	89.4

I-LoRA vs Baselines. We compare our I-LoRA strategy (at various allocation ratios) with standard LoRA and full fine-tuning regarding performance, trainable parameters, training time, and GPU memory footprint. As shown in Figure 8, I-LoRA drastically reduces training costs while matching the performance of standard LoRA and even surpassing full fine-tuning. This validates both the superiority and the necessity of our importance-aware design.

Visualization. To understand routing behavior, we visualize the average activation per encoder block in Figure 9. Activations are highly selective and not strictly depth-dependent, indicating that different frames rely on distinct semantic features across the network. This highlights the encoder’s modularity and the effectiveness of our recurrent dynamic routing. More visualizations are provided in the supplementary material, including per-frame block activations and frame-wise activation rates.

Table 8. **Generalizability of RDS on DINOv3.** RDS achieves a Pareto improvement, accelerating inference by up to $2.1\times$ while improving performance, indicating removal of detrimental backbone redundancy.

Model for video object segmentation	$\mathcal{J}\&\mathcal{F}$	FPS
DINOv3-L (distilled from DINOv3-7B)	38.4	2.7
+RDS _{0.6} (LoRA rank=32)	40.3	4.2 (1.6 $\times$)
+RDS _{0.5} (LoRA rank=32)	39.9	4.9 (1.8 $\times$)
+RDS _{0.4} (LoRA rank=32)	38.5	5.6 (2.1 $\times$)

Generalizability and orthogonality. To demonstrate that RDS is not limited to SAM2, we apply it to a distilled model. Since our method targets redundancy at the backbone block level, it is architecture-agnostic (generalizable beyond SAM2) and orthogonal to distillation. We verify this on DINOv3’s [33] distilled version (see Table 8).

5. Conclusion

In this paper, we aim to reduce the computational burden of SAM2 for video object segmentation and tracking. Observing that different samples favor different submodels, we propose Recurrent Dynamic Submodel. This framework features a Prediction-Aware Router that adaptively selects the optimal submodel for each frame by leveraging fused information from current and previous frames. To further reduce adaptation overhead, we introduce Importance-Aware LoRA, which assigns training weights only to the most critical components. Extensive experiments demonstrate a strong performance–speed trade-off, while requiring minimal trainable parameters and training data.

Limitations and Future Work. Our dynamic approach is currently implemented based on SAM2, but extending it to other vision foundation models, such as those designed for streaming video understanding [29, 39] and generation [19, 22], represents a promising direction. This could enable efficiency by dynamically allocating computational resources to targets across continuous frames.

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